Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible β-Sheet-like Order

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Abstract

Small-angle X-ray photon correlation spectroscopy (SA-XPCS) of the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$ separates dynamic arrest from the assembly that SAXS records. Four loadings held between 28.00 and 28.75 °C all assemble, yet only those above 28.00 °C arrest within the hold, and SAXS cannot tell their states apart: nearly the same profile is mobile at 28.00 °C and arresting a quarter of a degree higher. At 28.50 °C the fast-decay amplitude of a two-step correlation function falls from 0.80 to

0.07 at 175 nm within an hour, and less far at every smaller length scale down to 83 nm. The time to half arrest shortens fivefold over 0.5 °C, and rescaling the waiting time by $\exp(\Delta T/T_s)$ collapses all four onto one master curve with $T_s = 0.22 \pm 0.02$ K at every length scale measured. Temperature-dependent SAXS/WAXS resolves a reversible transition to a hierarchical assembly carrying β-sheet-like peaks at $Q = 1.38$ and 0.72 Å^{-1} that vanish on cooling.

Amyloid-like β-sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence homology are associated with Alzheimer's disease,[1] Parkinson's disease,[2] type II diabetes,[3] prion diseases,[4] and systemic light-chain amyloidosis.[5] Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.[6] Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules[7] and build the extracellular matrix of bacterial biofilms,[8] and related assemblies are increasingly used as mechanically robust biomaterials.[9] What differs across these systems is not whether β-sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,[10] and where a sequence falls in that range is a materials-design variable.

Reversible β-sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.[10–13] Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.[14–18] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.[17–19] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.[20,21] Although these studies establish the arrested state from time-averaged scattering and rheology, how the assembly evolves toward arrest remains elusive.

Whether an assembled state is still evolving is a question for a dynamical measurement. In colloidal gels, microscopic relaxation continues to evolve after the time-averaged structure has nearly stopped changing, a nonequilibrium process termed physical aging.[22,23] Related heterogeneous relaxation has been resolved during protein liquid–liquid phase separation,[24] protein gelation,[25] and cage relaxation in crowded protein solutions.[26] A protein-rich morphology can likewise keep reorganizing while its SAXS profile changes modestly, through junction exchange, whose kinetics set the network relaxation time in associative networks built from this same pentapeptide family,[27] and through the residual coarsening that follows arrested phase separation.[28,29] A dynamical measurement can also separate two routes to arrest. If the assembly stiffens as a whole, relaxation times lengthen continuously. If it instead converts region by region, the material that is still mobile keeps its relaxation rate while its share of the scattering falls. A bulk measurement averages the two together. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) probes the evolution directly through speckle decorrelation at the same low-Q length scales sampled by SAXS.

Here, temperature-dependent SAXS/WAXS and isothermal SA-XPCS provide complementary structural and dynamical views of concentrated aqueous $(\mathrm{VPAVG})_{30}$, an ELP built from 30 repeats of the pentapeptide valine-proline-alanine-valine-glycine. At 10 °C the scattering is dominated by molecular dimensions with no detectable mesoscale assembly; heating to 30 °C produces hierarchical low-Q scattering and high-Q peaks compatible with β-sheet-like packing, and cooling restores the low-temperature profile. In a separate experiment, isothermal SA-XPCS on four loadings held between 28.00 and 28.75 °C reveals a two-step decay whose fast fraction falls with elapsed time, first at the largest length scales. Arrest is set by temperature rather than by the structure SAXS records, one rescaling of time carries all four holds onto a single curve, and the local order that accompanies the assembly stays thermally reversible.

Figure 1 summarizes the SA-XPCS experiment. Its upper-left cartoon illustrates one candidate pathway, not a direct observation: protein-poor solvent regions appearing and growing within the slower protein-rich phase, coarsening it toward a network-like morphology. This dynamic-asymmetry picture follows Tanaka's viscoelastic phase-separation framework.[20,28] The $(VPAVG)_{30}$ sample was approximately $100\,\mathrm{mg\,mL^{-1}}$ in the buffer given below. Each 2 s SA-XPCS acquisition at Advanced Photon Source (APS) beamline 8-ID-I (Experimental Section) used a previously unexposed sample position to limit radiation damage (SI Section 6); temporal and azimuthal averaging yielded $I(Q)$ and multi-tau correlation analysis yielded $g_2(Q, \Delta t)$.

Figure 2 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were measured at APS 12-ID-B by simultaneous SAXS/WAXS at 13.3 keV.[30–32] The "Measurement" aliquot had undergone more than 10 heating/cooling cycles; the "Reference" aliquot had stayed at 4 °C. At 10 °C the two overlap, and a Guinier fit to the Reference gives an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S3), a dispersed state with no mesoscale aggregates down to $Q = 0.0081$ Å^{-1}, which corresponds to structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å^{-1}). Both also develop the same pair of high-Q features, at $Q = 1.38$ and 0.72 Å^{-1} in the Reference and 1.36 and 0.74 Å^{-1} in the Measurement. The cycled aliquot alone makes the point: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at $1\,^{\circ}\mathrm{C\,min^{-1}}$ (Figure S4). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on every heating and returned to the same low-temperature state on every cooling (SI Section 7). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle

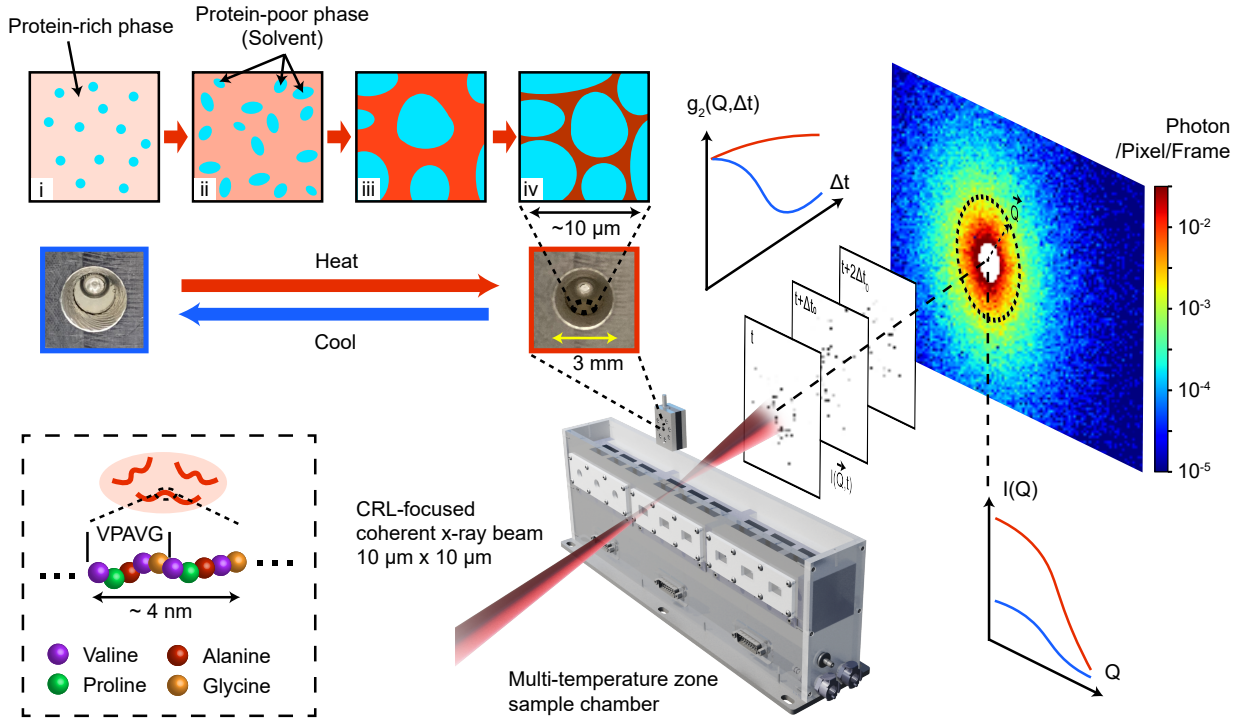


Figure 1: SA-XPCS measurement of thermally induced $(\mathrm{VPAVG})_{30}$ assembly at APS 8-ID-I. Each 100,000-frame set of sparse 20 µs frames feeds two analyses. Averaging them in time gives the SAXS intensity $I(Q)$: how much structure the sample has at each length scale $2\pi/Q$, which grows at low Q on heating. Correlating them in time gives $g_2(Q, \Delta t)$: how fast the structure at that same length scale rearranges. A decay in g_2 means the density fluctuations at $2\pi/Q$ are still moving; a flat g_2 means they are frozen over the accessible delay times Δt. Both are sketched beside the detector (red: assembled; blue: dispersed; illustrative, not data). Center: the aluminum cell, liquid chamber 3 mm across. The photos, outlined in blue and red, show the ELP sample below and above the transition. Lower left: one $(\mathrm{VPAVG})_{30}$ repeat, residues colored as keyed. Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; the graininess is sparse counting statistics, not resolved speckle. Upper left, stages i–iv: a conceptual illustration of the viscoelastic phase-separation pathway,[20,28] in which protein-poor solvent pools (cyan) nucleate and grow within the still-continuous protein-rich phase (deepening red: more protein), coarsening it toward a network-like morphology.

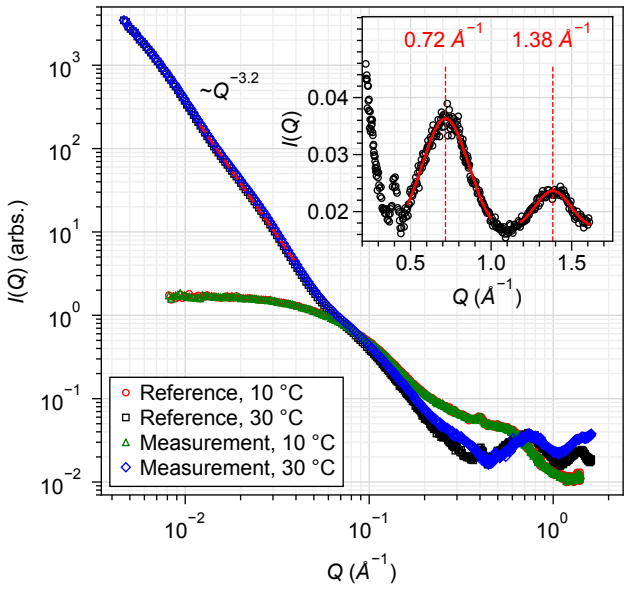


Figure 2: Temperature-dependent structure and reversibility of $(\mathrm{VPAVG})_{30}$. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S3). Heating produces an approximate $Q^{-3.2}$ low-Q response and peaks at $Q = 1.38$ and 0.72 Å^{-1}, corresponding to real-space spacings of approximately 4.5 and 8.8 Å, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040$ Å^{-1}. Inset: the high-Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å^{-1}, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

scattering signifies density heterogeneity extending across the probed interval rather than a single resolved structural scale.[33] Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.[20,21] The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross-β-like assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.[6,34] Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 Å^{-1} (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross-β crystallites. Their concurrent appearance with the low-Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β-sheet-like associations that are plausible, rather than uniquely established, physical junctions.

To resolve mesoscale arrest isothermally, four loadings were each held at 6 °C for more than 30 min, warmed to 20 °C, then taken at 0.1 °C min^{-1} to a final temperature between 28.00 and 28.75 °C and held there. The waiting time t_w starts at the first acquisition with the calibrated resistance-temperature detector (RTD) at the cell reading the target. Figure 3 shows the loading held at 28.50 °C; the other three are in SI Figures S7, S8 and S10, and Figure 4 puts all four on one axis. Each t_w labels a group of averaged consecutive acquisitions at previously unexposed positions (SI Section 8.1). The SAXS intensity starts close to the 6 °C reference of the same loading and grows fourfold over the hold at low Q, staying broad and monotonic with no peak emerging (Figure 3a), so the assembly gains density contrast without settling on a single characteristic domain size. Correlation functions are fitted in the five lowest Q bins, 0.00359 to 0.00756 Å^{-1} or 175 to 83 nm; above them the scattered intensity has fallen by more than an order of magnitude and g_2 is dominated by counting noise.

The intensity autocorrelation function g_2 for a single SA-XPCS acquisition is defined as:

$$g_2(Q, \Delta t) = \frac{\langle I(Q,t) I(Q, t+\Delta t)\rangle_t}{\langle I(Q,t)\rangle_t^2}, \tag{1}$$

where $I(Q,t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/Q$. At the largest length scale the correlation functions show a two-step decay: a millisecond-scale relaxation followed by an apparent plateau that persists through the 2 s acquisition (Figure 3b), the fast step being resolved only by the 50 kHz framing of the XSPA-500k.[35]

As t_w elapses over repeated acquisitions, the amplitude of the first decay decreases and the plateau rises. Such two-step relaxation and growth of a slow component are characteristic of heterogeneous kinetic arrest in protein and colloidal gels.[22,25,36] The two-step decays in g_2 were parameterized by:

$$\begin{aligned} g_2(Q, \Delta t, t_w) &= 1 + \beta \left| g_1(Q, \Delta t, t_w) \right|^2, \\ g_1(Q, \Delta t, t_w) &= f(Q, t_w) \exp\left[- \left(D_0(t_w)\, Q^2\, \Delta t \right)^{p_{\text{fast}}} \right] \\ &\quad + \left[1 - f(Q, t_w) \right] \exp\left[- \left(\frac{\Delta t}{\tau_{\text{slow}}(Q, t_w)} \right)^{p_{\text{slow}}(t_w)} \right], \end{aligned} \tag{2}$$

The first line is the Siegert relation, in which g_1 is the normalized intermediate scattering function and β is the instrumental speckle contrast, measured on a static reference and held fixed (SI Section 8.2). The second line is the two-mode model, where f is the fitted fast fraction and τ_{slow} an effective slow relaxation time. The effective fast relaxation time is replaced by $1/(D_0 Q^2)$ in the fast term, where D_0 is a fitted diffusion coefficient. That form carries a physical claim: the population that is still mobile moves diffusively, so a density fluctuation of wavelength $2\pi/Q$ relaxes in the time a scatterer needs to diffuse across it and the relaxation rate grows as Q^2. Released from the constraint the fitted exponent has a median of -2.5, and the collapse of Figure 4 survives releasing it (SI Section 5.2). The

stretching exponents p_{fast} and p_{slow} describe the breadth of the relaxation-time distributions; both are shared across the five Q bins of a group and both are free from one group to the next. A stretched correlation function can still be diffusive as long as p_{fast} does not depend on Q and the characteristic time scales with Q^{-2}. The rate of the fast mode is reported as D_0 itself, a parameter of the fit rather than a quantity derived from it. p_{fast} falls from about 1 while a hold is still mobile to about 0.2 once it has arrested, so a single value shared along a hold is rejected by an F test at $p < 10^{-16}$ in every hold that arrests. Holding p_{fast} at 1, which makes the fast mode a single exponential, describes the one hold that stays mobile but not the three that arrest, where the mean reduced χ^2 rises from 1.6 to between 7 and 10. The breadth of the fast relaxation is therefore a measured feature of arrest rather than a fitting convenience. A stretching exponent below 1 means the fast term is a distribution of relaxation times rather than a single rate, and the lower it falls the wider that distribution: the delay interval over which the fast term decays from 90 to 10% of its amplitude is $(\ln 10/\ln \frac{10}{9})^{1/p_{\mathrm{fast}}}$, which widens from 1.3 decades at $p_{\mathrm{fast}} = 1$ to 6.7 at 0.2. Late in a hold, the material that is still mobile relaxes across a very wide spread of time scales. τ_{slow} exceeds the acquisition window, so it is an extrapolated descriptor; the primary quantities are f and D_0.

Figure 3c shows the fitted fast fraction falling at every measured Q over the hold. At $t_w = 6\,\mathrm{s}$, f is 0.80 at $Q = 0.00359$ Å^{-1} (175 nm); by 3560 s that bin has fallen to 0.07, so the complementary arrested fraction $1 - f$ has gone from 0.20 to 0.93 in under an hour. At every analyzed time f also increases monotonically with Q, in all seven groups, so arrest develops first and most strongly at the largest length scale probed, 175 nm, and progressively less so toward 83 nm. That ordering holds within each group, so it reflects length scale rather than the trend with t_w, and it ties the growing low-Q SAXS structure to the arrest fraction measured on the very same frames. Note that XPCS does not locate the arrested regions, and the scattering contributions are weighted by structure and contrast, therefore $1 - f$ is not a geometric volume fraction.

Figure 3c also shows that what survives in the mobile population keeps moving. While

the sample is still largely mobile, D_0 falls from 8.0 to 2.3×10^7 Å^2 s^{-1}. Through the rest of the hold it varies between 0.9 and 1.7×10^7 Å^2 s^{-1} while $1 - f$ climbs from 0.46 to 0.93. Progressive stiffening would look different: as regions tightened, their density fluctuations would relax ever more slowly and would stop relaxing diffusively. Instead the surviving fraction keeps a diffusive Q^{-2} relaxation at a coefficient that varies by a factor of two through arrest while f falls by more than a factor of ten. The amplitude changes much further than the rate and does so monotonically, so at this stage of arrest it is the amplitude that carries the change. The same pairing of a plateauing fast time scale with a declining fast amplitude was found in a thermoreversible nanoparticle gel,[22] and the dynamic heterogeneity of such gels is established by confocal microscopy and simulation.[29,37]

The fast amplitude and the fast rate are not equally well determined late in a hold: f is set by the plateau height and stays measured, whereas any fast-mode time scale needs a resolved decay and is lost once f falls to a few percent. τ_{slow} lies beyond the acquisition window and is not used for scaling. Fresh material at every acquisition keeps cumulative dose from tracking t_w, so the trend in f cannot be a dose artifact, and a control at 23–41 times this flux shows no change in g_2 with dose rate (Figure S6).

The other three loadings show what sets the arrest (Figure 4, SI Section 8 and Figures S7, S8 and S10). All four assemble, their low-Q intensity rising on the ramp and again during the hold, but they arrest at very different rates: the time for f at the largest length scale to fall by half shortens fivefold between 28.25 and 28.75 °C. SAXS cannot tell these states apart. At the onset of arrest at 28.25 and 28.75 °C, $I(Q)$ lies within 18% of the final 28.00 °C profile at every Q, and that profile is still mobile after 6500 s. Arrest is therefore a step distinct from assembly, set by temperature rather than by the structure that SAXS records.

One time scale describes all four. Rescaling the waiting time by $\exp[(T - T_0)/T_s]$ collapses the four $f(t_w)$ curves onto a single master curve (Figure 4b), with $T_s = 0.22 \pm 0.02$ K, the mean over the five Q bins and the bootstrap uncertainty of collapsing one bin on its own. Collapsed one bin at a time it reads 0.21 to 0.24 K across the five Q values, from 175 to 83 nm

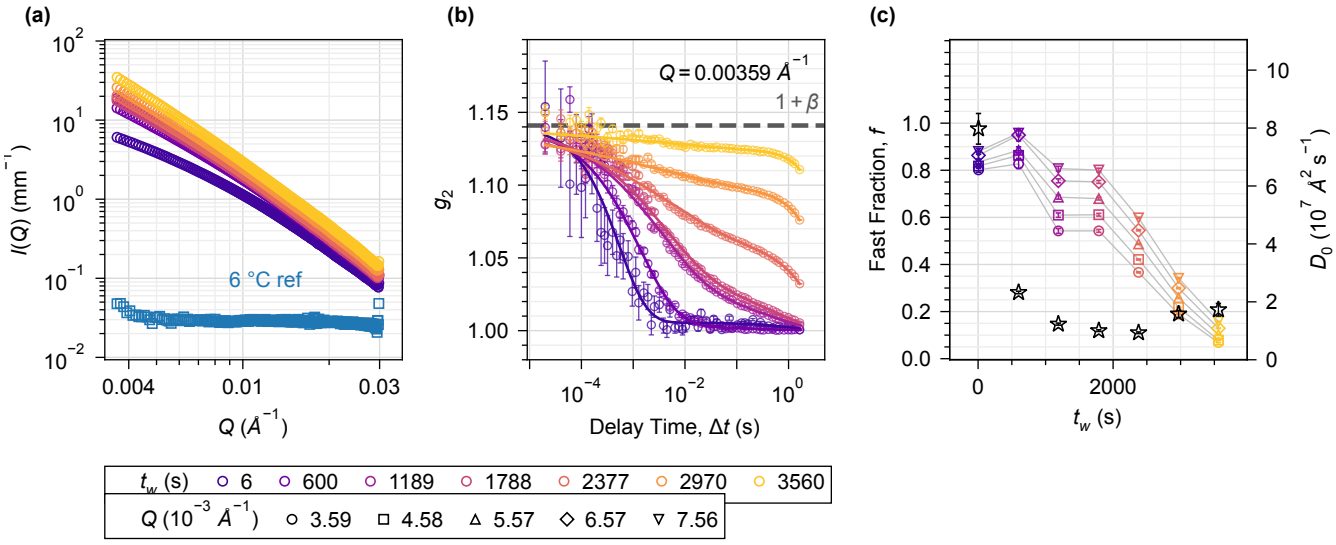


Figure 3: Assembly and dynamic arrest of $(\mathrm{VPAVG})_{30}$ held at 28.50 °C. Over the hold the mesoscale structure keeps growing while the dynamics stop: the low-Q intensity rises fourfold while the fast fraction falls from 0.80 to 0.07 at the largest length scale, so the two are separate events rather than two views of one. (a) Background-subtracted SAXS on the absolute scale of Figure S5, with the empty cell subtracted as in SI Section 8.2 and the 6 °C reference of the same loading as blue squares; profiles are drawn from the lowest fitted Q bin outward. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 at $\beta = 0.141$, p_{fast} free in each group; the dashed line is $1 + \beta$. (c) Left axis: the fast fraction f at the five Q values, marker shape denoting Q, grey lines guides to the eye. Right axis, black stars: the fitted diffusion coefficient D_0 of the fast mode. The amplitude falls more than tenfold while the rate stays within an order of magnitude. The four remaining Q values are in Figure S9. Error bars are one-standard-deviation fit uncertainties; colour denotes waiting time, keyed along the bottom.

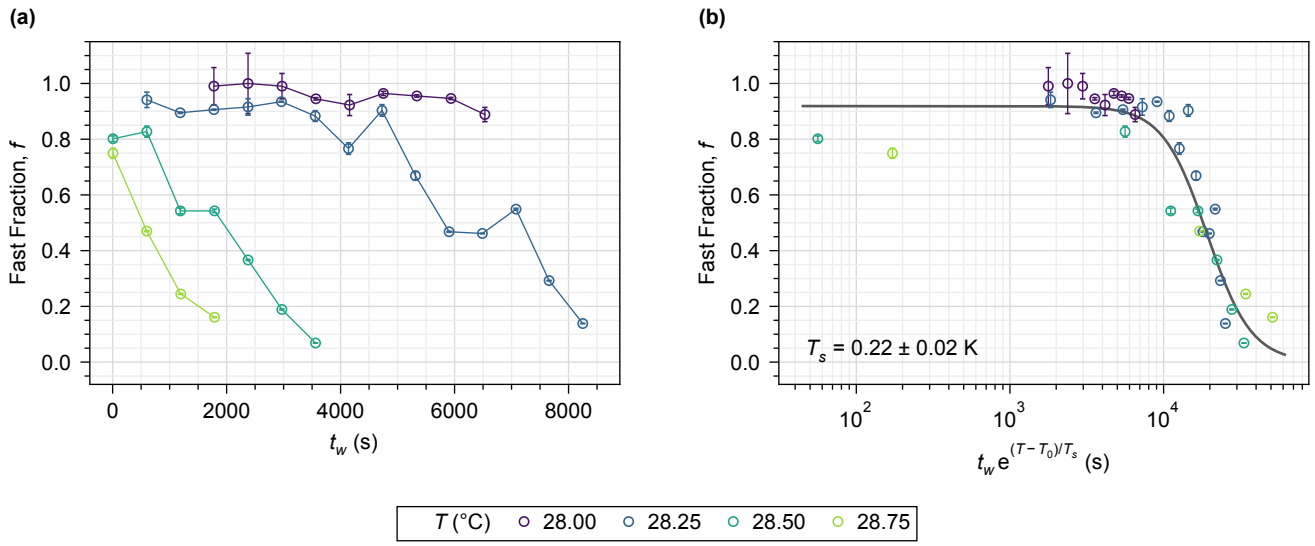


Figure 4: How fast arrest runs, and the single time scale that collapses it. (a) The fast fraction f at the largest length scale, $Q = 0.00359$ Å^{-1}, against t_w. At 28.00 °C f stays at 0.89 or above for 6500 s; above that temperature it falls, and sooner the warmer the hold. (b) The same points with time rescaled by $\exp[(T - T_0)/T_s]$, which brings all four onto one master curve: the grey line is a weighted fit of $f = A/[1 + (x/x_c)^n]$ to the pooled points with T_s held at the mean of the five values fitted one Q bin at a time, and rescaling improves the residual 2.4- to 3.0-fold at every Q. $T_0 = 28.00$ °C is a convention rather than a measured transition temperature and leaves T_s unchanged (SI Section 8). Collapsing each Q bin on its own returns 0.21 to 0.24 K from 175 to 83 nm (Figure S11i), a drift too small and the bins too correlated to call a trend. The collapse is approximate: the master curve lies about eight times the fit uncertainties from the points, and the first acquisitions of the 28.50 and 28.75 °C holds, leftmost in (b), fall below it because those loadings had already arrested partway during their longer ramps. Colour denotes temperature, keyed along the bottom; error bars are one-standard-deviation fit uncertainties.

(Figure S11i). That says more than that warmer is faster: the sample passes through the same sequence of dynamical states at every temperature and at every length scale measured, and temperature sets only the pace. The 28.00 °C loading, which never arrests inside its 6500 s hold, is not an exception but the same curve seen too early, the scaling placing its half-arrest near 15 600 s. A scaling temperature of a few tenths of a kelvin is itself the signature of a cooperative step. The same collapse was introduced for thermoreversible colloidal gels, where the formation rate varies with quench depth far more steeply than the pair attraction alone allows, and that steepness was attributed to the cooperative bonding needed to build locally favored structures.[23,38] Applied to that system it returns $T_s = 0.33$ and 0.45 K for two particle sizes, corresponding to local structures held by roughly twenty bonds.[22] A single contact cannot produce such steepness: one bond's energy changes far too little over half a degree. The temperature sensitivity of an individual contact here is not independently calibrated, so no bond number is extracted, but arrest being gated by something that must form many contacts at once is what the scaling requires.

This changes what is understood to pace the assembly. Above the LCST, ELP assembly is driven by hydrophobic desolvation, and desolvation is well under way before any hold begins: the scattered intensity has already grown two- to fourfold by 27.5 °C on the ramp (SI Section 8). What the scaling adds is that desolvation is not the step that sets the clock. The equilibrium of a single contact shifts very little over half a degree, yet half a degree changes the rate of arrest by a factor of five, so the rate-limiting event is the concerted formation of a junction in which many contacts close together. The two dynamical observables describe that event from opposite sides. T_s measures how steeply its rate depends on temperature. p_{fast}, falling to about 0.2, measures how widely the relaxation times of the material that has not yet converted are spread, which is what a population distributed across many stages of junction assembly produces. Read together they place arrest as a region-by-region conversion, each region waiting on a step that many contacts must complete at once.

The same cooperativity carries to the reverse direction. A junction that needs many

contacts to form needs many to break before it comes apart, which is the condition for a thermal transition that is sharp and that repeats, and the structure does return through seven heating and cooling cycles (Figure S4). It also sets a practical scale. At $T_s = 0.22$ K, holding this sequence a quarter of a degree warmer shortens the time to arrest by about a factor of three, so the assembly rate of an ELP gel is adjustable over orders of magnitude by sub-degree temperature control, with the arrested state reached along the same pathway at every setting.

The growth of low-Q SAXS intensity and the decline of the fast fraction are two aspects of network formation, and the temperature series shows that they are distinct. Their sequence, assembly first and arrest after, is consistent with network-forming arrested phase separation in a dynamically asymmetric protein–solvent mixture, where concentration fluctuations build a connected dense phase and viscoelastic constraints suppress relaxation.[24,25,28,39] Related ELPs form bicontinuous networks with stress-relaxation times beyond 10^3 s.[20,21] The present measurements add a local structural element from a separate experiment: at high temperature $(\mathrm{VPAVG})_{30}$ carries thermally reversible β-sheet-like contacts. The separate experiments cannot place those contacts at the junctions, but the two are compatible in mechanism and the scaling temperature makes the link quantitative. Arrest that empties the mobile population rather than slowing it is what cooperative junctions produce, a region leaving that population once enough contacts form together; and β-sheet assembly is cooperative in exactly that sense, clearing a nucleation barrier before growth begins,[40] accelerating once started,[41] and extending by templated addition onto a self-complementary interface.[6]

The route to arrest matters for what a partially arrested network can do. A medium that stiffens throughout slows every solute within it. One that converts region by region keeps pockets that are not merely less viscous but still diffusive, so transport should follow the connectivity of those pockets rather than a bulk viscosity, and release should hold and then fall at a threshold rather than tail off. Because the arrested fraction evolves in time and can be measured, a network interrupted at a chosen waiting time carries a defined fraction,

a handle that neither composition nor the SAXS profile gives. The same question is open for biomolecular condensates, whose maturation from liquid to solid governs their function and their role in aggregation.[42,43] If maturation proceeds the same way, a partially matured condensate keeps interior regions that still exchange, and the liquid-to-solid transition is a connectivity threshold rather than a gradual drift. Condensates are too small and too dilute for the measurement reported here; a concentrated, sequence-defined mimic is not, and in this one arrest is local.

Four limitations bound the interpretation, and each points to the measurement that would lift it. First, the β-sheet-like assignment rests on WAXS peak positions rather than spectroscopy; attenuated-total-reflection infrared through a thermal cycle is the natural test, circular dichroism being poorly suited to a sample this turbid. Second, the high-Q ordering and the low-Q dynamics come from separate experiments, so no time-resolved causal relation between them is established; simultaneous WAXS and SA-XPCS on one loading would settle it. Third, the 2×10^{-5}–1.6 s delay range quantifies the fraction arrested on that scale rather than the infinite-time nonergodicity factor, for which linear shear rheology on a scaled-up preparation is the complement. Fourth, each temperature rests on a single loading, so the quoted uncertainties are fit uncertainties rather than between-sample variability, and the scaling rests on three arresting temperatures spanning half a degree. The collapse is approximate: rescaling brings the loadings threefold closer together but not to within the measurement uncertainty, and T_s drifts by 13% across the five Q bins. More temperatures, with replicate loadings at each, would settle both. T_0 is a convention and not a measured transition temperature, so no reference time is quoted from it.

Subject to those constraints, the two measurements give one picture. SA-XPCS resolves what SAXS cannot: whether an assembly has arrested. Nearly the same mesoscale structure is mobile at 28.00 °C and arresting a quarter of a degree higher, arrest reaches the largest length scales first, and once under way it advances by converting mobile material rather than by slowing it. One rescaling of time carries every hold onto the same curve, so temperature

sets the pace of a single pathway. SAXS/WAXS resolves what the assembly is made of: a hierarchical network carrying β-sheet-like local order that vanishes on cooling. Put together, a protein network can arrest region by region while the contacts that build it stay thermally erasable, which is what makes amyloid-like packing a candidate switchable junction rather than a commitment to permanent aggregation.

Experimental Section. The Sanger-verified construct encoded an N-terminal His-tagged $(VPAVG)_{30}$ protein (complete sequence in the Supporting Information). Recombinant $(VPAVG)_{30}$ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L^{-1} HEPES, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (Figure S1b). Protein concentration was determined from A_{280} using the sequence-specific extinction coefficient (ε_{280} = 1490 M^{-1} cm^{-1}, 15,461 Da construct); the final concentration was approximately 100 mg mL^{-1}.

Temperature-dependent SAXS/WAXS measurements were performed at APS 12-ID-B with a 13.3 keV beam of approximately 100×150 μm^2 and an incident flux of approximately 5×10^{11} photons s^{-1}. At each set point the capillary was held at temperature for at least 10 min before exposure. SAXS and WAXS were acquired simultaneously in 0.1 s exposures with an Eiger2 9M at approximately 2 m and a Pilatus 300K at 0.42 m. SA-XPCS was performed at APS 8-ID-I with a 10.94 keV, 10×10 μm^2 partially coherent beam and a Rigaku XSPA-500k detector 7.8 m downstream. Each acquisition contained 100,000 frames at 50 kHz (20 µs per frame, 2 s total), and the four holds were measured in one beamtime in a single cell block. Further acquisition, fitting and radiation-control details are in the Supporting Information.

Supporting Information

Recombinant synthesis and purification; gene-construction sequences and SDS-PAGE analysis; sample environments; SAXS/WAXS calibration; absolute scattering cross section; SA-XPCS analysis and fitting; speckle-contrast calibration; radiation controls; Guinier analysis; temperature dependence of arrest; the scaling collapse at every measured Q; thermal-cycling repeatability; a still frame and link for the video of the reversible thermal transition; code and reduced-data availability; and AI-assisted development (PDF).

Data and Code Availability

Reduced data and analysis/figure-generation code are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`. The repository includes the 8-ID-I and 12-ID-B datasets used for the figures, radiation-control data, and analysis scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

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AI-assisted manuscript and code preparation. The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and with development of the data-

analysis and figure-generation code. The Claude models were accessed through Argo, Argonne National Laboratory's internal generative AI interface (see Supporting Information). That code and the reduced data it operates on are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Conflict of Interest

The authors declare no competing financial interest.

References

(1) Masters, C. L.; Simms, G.; Weinman, N. A.; Multhaup, G.; McDonald, B. L.; Beyreuther, K. Amyloid Plaque Core Protein in Alzheimer Disease and Down Syndrome. *Proceedings of the National Academy of Sciences of the United States of America* **1985**, *82*, 4245–4249.

(2) Serpell, L. C.; Berriman, J.; Jakes, R.; Goedert, M.; Crowther, R. A. Fiber Diffraction of Synthetic α-Synuclein Filaments Shows Amyloid-Like Cross-β Conformation. *Proceedings of the National Academy of Sciences of the United States of America* **2000**, *97*, 4897–4902.

(3) Cooper, G. J. S.; Willis, A. C.; Clark, A.; Turner, R. C.; Sim, R. B.; Reid, K. B. M. Purification and Characterization of a Peptide from Amyloid-Rich Pancreases of Type 2 Diabetic Patients. *Proceedings of the National Academy of Sciences of the United States of America* **1987**, *84*, 8628–8632.

(4) Pan, K.-M.; Baldwin, M.; Nguyen, J.; Gasset, M.; Serban, A.; Groth, D.; Mehlhorn, I.; Huang, Z.; Fletterick, R. J.; Cohen, F. E.; Prusiner, S. B. Conversion of Alpha-Helices

into Beta-Sheets Features in the Formation of the Scrapie Prion Proteins. *Proceedings of the National Academy of Sciences of the United States of America* **1993**, *90*, 10962–10966.

(5) Radamaker, L.; Lin, Y.-H.; Annamalai, K.; Huhn, S.; Hegenbart, U.; Schönland, S. O.; Fritz, G.; Schmidt, M.; Fändrich, M. Cryo-EM Structure of a Light Chain-Derived Amyloid Fibril from a Patient with Systemic AL Amyloidosis. *Nature Communications* **2019**, *10*, 1103.

(6) Sawaya, M. R.; Sambashivan, S.; Nelson, R.; Ivanova, M. I.; Sievers, S. A.; Apostol, M. I.; Thompson, M. J.; Balbirnie, M.; Wiltzius, J. J. W.; McFarlane, H. T.; Madsen, A. Ø.; Riekel, C.; Eisenberg, D. Atomic Structures of Amyloid Cross-β Spines Reveal Varied Steric Zippers. *Nature* **2007**, *447*, 453–457.

(7) Maji, S. K.; Perrin, M. H.; Sawaya, M. R.; Jessberger, S.; Vadodaria, K.; Rissman, R. A.; Singru, P. S.; Nilsson, K. P. R.; Simon, R.; Schubert, D.; Eisenberg, D.; Rivier, J.; Sawchenko, P.; Vale, W.; Riek, R. Functional Amyloids as Natural Storage of Peptide Hormones in Pituitary Secretory Granules. *Science* **2009**, *325*, 328–332.

(8) Chapman, M. R.; Robinson, L. S.; Pinkner, J. S.; Roth, R.; Heuser, J.; Hammar, M.; Normark, S.; Hultgren, S. J. Role of Escherichia coli Curli Operons in Directing Amyloid Fiber Formation. *Science* **2002**, *295*, 851–855.

(9) Knowles, T. P. J.; Mezzenga, R. Amyloid Fibrils as Building Blocks for Natural and Artificial Functional Materials. *Advanced Materials* **2016**, *28*, 6546–6561.

(10) Pálmadóttir, T.; Getachew, J.; Ortigosa-Pascual, L.; Axell, E.; Wei, J.; Olsson, U.; Knowles, T. P. J.; Linse, S. On the Reversibility of Amyloid Fibril Formation. *Biophysics Reviews* **2025**, *6*, 011303.

(11) Kato, M. et al. Cell-free Formation of RNA Granules: Low Complexity Sequence Domains Form Dynamic Fibers within Hydrogels. *Cell* **2012**, *149*, 753–767.

(12) Hughes, M. P.; Sawaya, M. R.; Boyer, D. R.; Goldschmidt, L.; Rodriguez, J. A.; Cascio, D.; Chong, L.; Gonen, T.; Eisenberg, D. S. Atomic structures of low-complexity protein segments reveal kinked beta sheets that assemble networks. *Science* **2018**, *359*, 698–701.

(13) Luo, F.; Gui, X.; Zhou, H.; Gu, J.; Li, Y.; Liu, X.; Zhao, M.; Li, D.; Li, X.; Liu, C. Atomic structures of FUS LC domain segments reveal bases for reversible amyloid fibril formation. *Nature Structural & Molecular Biology* **2018**, *25*, 341–346.

(14) Meyer, D. E.; Chilkoti, A. Quantification of the Effects of Chain Length and Concentration on the Thermal Behavior of Elastin-like Polypeptides. *Biomacromolecules* **2004**, *5*, 846–851.

(15) Amiram, M.; Quiroz, F. G.; Callahan, D. J.; Chilkoti, A. A highly parallel method for synthesizing DNA repeats enables the discovery of 'smart' protein polymers. *Nat. Mater.* **2011**, *10*, 141–148.

(16) Quiroz, F. G.; Chilkoti, A. Sequence heuristics to encode phase behaviour in intrinsically disordered protein polymers. *Nature Materials* **2015**, *14*, 1164–1171.

(17) Varanko, A. K.; Su, J. C.; Chilkoti, A. Elastin-Like Polypeptides for Biomedical Applications. *Annu. Rev. Biomed. Eng.* **2020**, *22*, 343–369.

(18) Saha, S.; Banskota, S.; Roberts, S.; Kirmani, N.; Chilkoti, A. Engineering the architecture of elastin-like polypeptides: From unimers to hierarchical self-assembly. *Adv. Ther.* **2020**, *3*, 1900164.

(19) Sing, M. K.; Glassman, M. J.; Vronay-Ruggles, X. T.; Burghardt, W. R.; Olsen, B. D. Structure and rheology of dual-associative protein hydrogels under nonlinear shear flow. *Soft Matter* **2017**, *13*, 8511–8524.

(20) Glassman, M. J.; Olsen, B. D. Arrested Phase Separation of Elastin-like Polypeptide Solutions Yields Stiff, Thermoresponsive Gels. *Biomacromolecules* **2015**, *16*, 3762–3773.

(21) Glassman, M. J.; Avery, R. K.; Khademhosseini, A.; Olsen, B. D. Toughening of Thermoresponsive Arrested Networks of Elastin-Like Polypeptides To Engineer Cytocompatible Tissue Scaffolds. *Biomacromolecules* **2016**, *17*, 415–426.

(22) Bahadur, D.; Zhang, Q.; Dufresne, E. M.; Grybos, P.; Kmon, P.; Leheny, R. L.; Maj, P.; Narayanan, S.; Szczygiel, R.; Swan, J. W.; Sandy, A.; Ramakrishnan, S. Evolution of structure and dynamics of thermo-reversible nanoparticle gels—A combined XPCS and rheology study. *J. Chem. Phys.* **2019**, *151*, 104902.

(23) Chen, Y.; Zhang, Q.; Ramakrishnan, S.; Leheny, R. L. Memory in aging colloidal gels with time-varying attraction. *J. Chem. Phys.* **2023**, *158*, 024906.

(24) Girelli, A.; Rahmann, H.; Begam, N.; Ragulskaya, A.; Reiser, M.; Chandran, S.; Westermeier, F.; Sprung, M.; Zhang, F.; Gutt, C.; Schreiber, F. Microscopic Dynamics of Liquid-Liquid Phase Separation and Domain Coarsening in a Protein Solution Revealed by X-Ray Photon Correlation Spectroscopy. *Physical Review Letters* **2021**, *126*, 138004.

(25) Begam, N.; Ragulskaya, A.; Girelli, A.; Rahmann, H.; Chandran, S.; Westermeier, F.; Reiser, M.; Sprung, M.; Zhang, F.; Gutt, C.; Schreiber, F. Kinetics of Network Formation and Heterogeneous Dynamics of an Egg White Gel Revealed by Coherent X-Ray Scattering. *Physical Review Letters* **2021**, *126*, 098001.

(26) Chushkin, Y.; Gulotta, A.; Roosen-Runge, F.; Pal, A.; Stradner, A.; Schurtenberger, P. Probing Cage Relaxation in Concentrated Protein Solutions by X-Ray Photon Correlation Spectroscopy. *Physical Review Letters* **2022**, *129*, 238001.

(27) Sing, M. K.; Burghardt, W. R.; Olsen, B. D. Influence of End-Block Dynamics on Deformation Behavior of Thermoresponsive Elastin-like Polypeptide Hydrogels. *Macromolecules* **2018**, *51*, 2951–2960.

(28) Tanaka, H. Viscoelastic phase separation. *J. Phys. Condens. Matter* **2000**, *12*, R207–R264.

(29) Zhang, I.; Patrick Royall, C.; Faers, M. A.; Bartlett, P. Phase separation dynamics in colloid–polymer mixtures: the effect of interaction range. *Soft Matter* **2013**, *9*, 2076–2084.

(30) Advanced Photon Source 12-ID-B Home Page and Detector Configuration. Advanced Photon Source, Argonne National Laboratory, 2026; `https://www.aps.anl.gov/Sector-12/12-ID-B`, Accessed July 22, 2026.

(31) Xu, L.; Wang, Z.; Chen, J.; Chen, S.; Yang, W.; Ren, Y.; Zuo, X.; Zeng, J.; Wu, Q.; Sheng, H. Folded Network and Structural Transition in Molten Tin. *Nature Communications* **2022**, *13*, 126.

(32) Zheng, C. Y.; Yao, Y.; Deng, J.; Seifert, S.; Wong, A. M.; Lee, B.; Mirkin, C. A. Confined Growth of DNA-Assembled Superlattice Films. *ACS Nano* **2022**, *16*, 4813–4822.

(33) Teixeira, J. Small-Angle Scattering by Fractal Systems. *Journal of Applied Crystallography* **1988**, *21*, 781–785.

(34) Sunde, M.; Serpell, L. C.; Bartlam, M.; Fraser, P. E.; Pepys, M. B.; Blake, C. C. F. Common Core Structure of Amyloid Fibrils by Synchrotron X-Ray Diffraction. *Journal of Molecular Biology* **1997**, *273*, 729–739.

(35) Nakaye, Y.; Sakumura, T.; Sakuma, Y.; Mikusu, S.; Dawiec, A.; Orsini, F.; Grybos, P.; Szczygiel, R.; Maj, P.; Ferrara, J. D.; Taguchi, T. Characterization and performance evaluation of the XSPA-500k detector using synchrotron radiation. *Journal of Synchrotron Radiation* **2021**, *28*, 439–447.

(36) Chen, Y.; Rogers, S. A.; Narayanan, S.; Harden, J. L.; Leheny, R. L. Microscopic

ergodicity breaking governs the emergence and evolution of elasticity in glass-forming nanoclay suspensions. *Phys. Rev. E.* **2020**, *102*, 042619.

(37) Patrick Royall, C.; Williams, S. R.; Ohtsuka, T.; Tanaka, H. Direct observation of a local structural mechanism for dynamic arrest. *Nat. Mater.* **2008**, *7*, 556–561.

(38) Zhang, Q.; Bahadur, D.; Dufresne, E. M.; Grybos, P.; Kmon, P.; Leheny, R. L.; Maj, P.; Narayanan, S.; Szczygiel, R.; Ramakrishnan, S.; Sandy, A. Dynamic Scaling of Colloidal Gel Formation at Intermediate Concentrations. *Phys. Rev. Lett.* **2017**, *119*, 178006.

(39) Lu, P. J.; Zaccarelli, E.; Ciulla, F.; Schofield, A. B.; Sciortino, F.; Weitz, D. A. Gelation of Particles with Short-Range Attraction. *Nature* **2008**, *453*, 499–503.

(40) Ferrone, F. Analysis of Protein Aggregation Kinetics. *Methods in Enzymology* **1999**, *309*, 256.

(41) Knowles, T. P. J.; Waudby, C. A.; Devlin, G. L.; Cohen, S. I. A.; Aguzzi, A.; Vendruscolo, M.; Terentjev, E. M.; Welland, M. E.; Dobson, C. M. An Analytical Solution to the Kinetics of Breakable Filament Assembly. *Science* **2009**, *326*, 1533–1537.

(42) Banani, S. F.; Lee, H. O.; Hyman, A. A.; Rosen, M. K. Biomolecular condensates: organizers of cellular biochemistry. *Nat. Rev. Mol. Cell Biol.* **2017**, *18*, 285–298.

(43) Shin, Y.; Brangwynne, C. P. Liquid phase condensation in cell physiology and disease. *Science* **2017**, *357*.